

**Kathmandu University**  
**Department of Computer Science and Engineering**  
**Dhulikhel, Kavre**



**A Project Proposal**  
**on**  
**“Speech-Based Mental Health Screening Using Acoustic Features and LLMs”**

**[Code No: COMP 207]**  
**(For partial fulfillment of II Year I Semester in Computer Engineering)**

**Submitted by:**

**Abhyudaya Pokhrel (Reg. No. 037993-24)**

**Dalton Khatri (Reg. No. 037979-24)**

**Sahaj Wagle (Reg. No. 038012-24)**

**Saksham Dallakoti (Reg. No. 037965-24)**

**Submitted to:**

**Suman Shrestha**

**Department of Computer Science and Engineering**

**Submission Date:**

## **Abstract**

Mental health disorders such as depression, anxiety, and chronic stress affect millions of people worldwide and often remain undetected due to limited access to professional assessment and social stigma associated with seeking help. Advances in artificial intelligence have created opportunities for developing non-invasive systems that can analyze human communication patterns and provide early indicators of mental well-being. Speech and language are particularly valuable sources of information because they contain both linguistic content and behavioral cues that may reflect an individual's psychological state. However, relying solely on either text or speech can limit the effectiveness of such systems, as important information from the other modality may be overlooked. To address this challenge, this project proposes a multimodal mental health screening and support system that analyzes both spoken and textual user input. The system utilizes speech processing and natural language analysis techniques to extract meaningful patterns from user communication and combines information from multiple modalities to improve screening performance. In addition to generating mental health risk assessments, the system incorporates an explainability component that provides understandable reasons behind its predictions and a Large Language Model (LLM) module that generates supportive and empathetic responses. The proposed approach aims to create an accessible and practical tool for preliminary mental health screening while emphasizing that it is not intended to replace professional diagnosis or treatment. The system will be evaluated using standard classification metrics on publicly available mental health datasets to assess its effectiveness and reliability.

**Keywords:** Mental Health Screening, Multimodal Learning, Speech Analysis, Text Analysis, Artificial Intelligence, Large Language Models, Explainable AI

# Table of Contents

|                                                                                                                        |           |
|------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Abstract.....</b>                                                                                                   | <b>i</b>  |
| <b>Table of Contents .....</b>                                                                                         | <b>ii</b> |
| <b>List of Figures.....</b>                                                                                            | <b>iv</b> |
| <b>Acronyms/Abbreviations.....</b>                                                                                     | <b>v</b>  |
| <b>Chapter 1 Introduction.....</b>                                                                                     | <b>1</b>  |
| 1.1 Background .....                                                                                                   | 2         |
| 1.2 Motivation .....                                                                                                   | 3         |
| 1.3 Objectives.....                                                                                                    | 3         |
| 1.4 Expected Outcomes.....                                                                                             | 4         |
| 1.5 Significance of the Project .....                                                                                  | 4         |
| <b>Chapter 2 Related Works.....</b>                                                                                    | <b>6</b>  |
| 2.1 Multi-Modal Deep Learning System for Depression and Anxiety Detection                                              | 6         |
| 2.2 AI-Powered Mental Health Assessment Using Speech and Text Analysis ...                                             | 6         |
| 2.3 Mental Health Monitoring from Speech and Language.....                                                             | 7         |
| 2.4 Multimodal and Multiresolution Depression Detection from Speech and<br>Facial Landmark Features .....              | 7         |
| 2.5 Leveraging Audio and Text Modalities in Mental Health: A Study of LLM<br>Performance .....                         | 8         |
| 2.6 When LLMs Meet Acoustic Landmarks: Integrating Speech into Large<br>Language Models for Depression Detection ..... | 8         |
| <b>Chapter 3 Procedure and Methods .....</b>                                                                           | <b>10</b> |
| 3.1 Simplified Workflow of the Project .....                                                                           | 11        |
| <b>Chapter 4. System Requirement Specifications .....</b>                                                              | <b>13</b> |
| 4.1 Software Specifications.....                                                                                       | 13        |
| 4.1.1 Functional Requirements.....                                                                                     | 13        |
| 4.1.2 Non-Functional Requirements.....                                                                                 | 14        |
| 4.2 Hardware Specifications .....                                                                                      | 15        |
| <b>Chapter 5. Project Planning and Scheduling .....</b>                                                                | <b>16</b> |
| 5.1 Requirement Analysis and Planning (Week 1–2) .....                                                                 | 16        |
| 5.2 System Design (Week 3–4).....                                                                                      | 16        |

|                                                           |    |
|-----------------------------------------------------------|----|
| 5.3 Model Development (Week 5–7).....                     | 16 |
| 5.4 Integration and Interface Development (Week 7–9)..... | 17 |
| 5.5 Testing and Evaluation (Week 10–11).....              | 17 |
| <b>References</b> .....                                   | 18 |

## List of Figures

|                             |    |
|-----------------------------|----|
| Figure 1: Gantt Chart ..... | 17 |
|-----------------------------|----|

## **Acronyms/Abbreviations**

|          |                                                   |
|----------|---------------------------------------------------|
| AI       | Artificial Intelligence                           |
| AVEC     | Audio/Visual Emotion Challenge                    |
| DAIC     | Distress Analysis Interview Corpus                |
| DAIC-WOZ | Distress Analysis Interview Corpus – Wizard of Oz |
| F1       | F1 Score                                          |
| GB       | Gigabyte                                          |
| GPU      | Graphics Processing Unit                          |
| LLM      | Large Language Model                              |
| MFCC     | Mel-Frequency Cepstral Coefficients               |
| NLP      | Natural Language Processing                       |
| PHQ-8    | Patient Health Questionnaire-8                    |
| PTSD     | Post-Traumatic Stress Disorder                    |
| RAM      | Random Access Memory                              |
| SMM      | Speech, Music and Mind                            |
| UI       | User Interface                                    |

## Chapter 1 Introduction

Mental health disorders such as depression, anxiety, and chronic stress affect millions of individuals worldwide and represent a growing public health concern. Early identification of mental health challenges is important because timely intervention can improve outcomes and quality of life. Traditional assessment methods primarily rely on self-report questionnaires and clinical interviews, which may not always be accessible due to geographical, financial, or social barriers. Recent advances in artificial intelligence have created opportunities to develop intelligent systems that can assist in mental health screening through the analysis of natural human communication (Zhang et al., 2022).

Human communication conveys more than the literal meaning of words. The language people use, as well as the way they speak, often contains valuable indicators of their emotional and psychological condition. Research has shown that linguistic patterns, vocal characteristics, pauses, speaking rate, and acoustic variations can provide meaningful insights into mental well-being (Aloshban et al., 2021). As speech and text become increasingly important forms of human-computer interaction, there is growing interest in systems capable of analysing both modalities to support mental health assessment.

Recent studies have demonstrated that multimodal approaches combining speech and text generally outperform systems that rely on a single modality. While textual information captures what a person says, speech signals provide information about how it is said, including vocal cues that may reflect psychological distress (Nasir et al., 2016; Aloshban et al., 2021). Furthermore, the emergence of Large Language Models (LLMs) has enabled the generation of more natural explanations and supportive responses, creating opportunities for user-centered mental health support systems (Zhang et al., 2024).

This project proposes a multimodal mental health screening and support system that analyses both speech and text inputs to identify potential indicators of mental health concerns. The system aims to combine linguistic and acoustic information, provide

explainable screening results, and generate supportive feedback through artificial intelligence techniques. The proposed solution is intended to serve as an assistive screening tool and is not designed to replace professional diagnosis or clinical treatment.

## **1.1 Background**

The application of computational techniques to mental health assessment has evolved significantly over the past decade. Early systems relied primarily on manually engineered linguistic and acoustic features combined with conventional machine learning algorithms. Although these approaches demonstrated promising results, they often struggled to generalize across diverse populations and real-world environments.

With the advancement of deep learning, researchers began utilizing neural networks capable of automatically learning complex patterns from speech and text data. These models improved the ability to identify mental health indicators by capturing semantic, contextual, and behavioural characteristics that were difficult to model manually. According to Zhang et al. (2022), deep learning has become the dominant paradigm in mental illness detection, consistently outperforming many traditional feature-engineering approaches.

More recently, multimodal systems have emerged as a promising direction for mental health screening. Studies have shown that combining linguistic information with acoustic speech characteristics can improve classification performance because mental health conditions may manifest differently across individuals (Aloshban et al., 2021). Some individuals exhibit stronger linguistic indicators, while others display more prominent acoustic patterns such as reduced pitch variation, increased pauses, or changes in speaking energy. Research utilizing datasets such as DAIC-WOZ has further demonstrated the value of integrating multiple behavioural signals for depression assessment (Nasir et al., 2016).

In parallel, Large Language Models have introduced new possibilities for generating explanations and supportive responses based on analyzed information.



Recent studies indicate that LLM-based systems can effectively utilize textual and speech-derived information to support mental health screening and interpretation tasks (Ali et al., 2024; Zhang et al., 2024).

## **1.2 Motivation**

The motivation for this project stems from the growing need for accessible mental health support tools and the increasing capabilities of artificial intelligence technologies. Many individuals experiencing mental health difficulties do not seek professional assistance due to stigma, limited resources, or lack of access to qualified professionals. Consequently, there is a demand for systems capable of providing preliminary screening and encouraging users to seek further support when appropriate.

Research has shown that both speech and language contain important indicators of mental well-being. However, many existing systems focus exclusively on either textual information or speech characteristics, potentially overlooking complementary information available in the other modality (Aloshban et al., 2021). Recent multimodal studies have demonstrated that combining linguistic and acoustic features can improve the identification of mental health conditions and provide more reliable assessments (Diep et al., 2022).

Another motivating factor is the increasing availability of advanced language models capable of producing natural and context-aware responses. Rather than presenting users with only a prediction score, modern systems can generate understandable explanations and supportive feedback that enhance user engagement and transparency. This project aims to explore how these capabilities can be integrated into a unified framework for mental health screening and support.

## **1.3 Objectives**

The main objectives of this project are:

1. To develop a multimodal mental health screening system capable of processing both speech and text inputs.
2. To analyse linguistic and acoustic characteristics of user communication to identify potential indicators of mental health conditions.
3. To combine information from speech and text modalities to generate explainable mental health screening results.
4. To design and implement a user-friendly system that provides supportive feedback and insights through artificial intelligence techniques.

## **1.4 Expected Outcomes**

By completion of the project, we expect to deliver:

1. A functional multimodal mental health screening system capable of accepting both speech and text inputs for analysis.
2. A trained screening model that utilizes linguistic and acoustic information to identify potential mental health indicators and generate assessment results.
3. An explainability component that provides understandable insights into the factors contributing to the system's predictions.
4. A user-friendly application that delivers screening results along with supportive and informative feedback generated through artificial intelligence.

## **1.5 Significance of the Project**

This project lies at the intersection of artificial intelligence, speech processing, natural language processing, and mental health informatics. By combining multiple forms of communication within a single framework, it contributes to the growing body of research on multimodal mental health assessment.

Existing studies have demonstrated the effectiveness of combining speech and text information for identifying mental health conditions (Nasir et al., 2016; Alosban

et al., 2021; Diep et al., 2022). However, many systems remain focused primarily on classification performance and provide limited explanation or user interaction. The proposed project seeks to address this gap by integrating screening, explainability, and supportive response generation into a unified system.

More broadly, the project contributes toward the development of accessible digital tools that can assist in raising awareness of mental health concerns and encourage individuals to seek professional support when necessary. While not intended to replace clinical diagnosis, the system has the potential to serve as a valuable supplementary resource for preliminary mental health assessment.

## **Chapter 2 Related Works**

A number of prior studies have investigated the use of speech, language, and multimodal artificial intelligence techniques for automated mental health assessment. These studies provide valuable insights into the extraction of behavioural indicators from human communication and the integration of multiple modalities for depression and anxiety screening.

### **2.1 Multi-Modal Deep Learning System for Depression and Anxiety Detection**

Diep et al. (2022) designed a multimodal classification framework that combines deep audio representations, natural language text features, and clinically informed hand-crafted biomarkers for depression and anxiety detection. Their architecture extended the AudiBERT framework by integrating acoustic information, linguistic content, and domain-specific mental health indicators derived from established clinical screening tools. The study demonstrated that combining deep-learned representations with clinically meaningful features improved screening performance, increasing F1-scores for both depression and anxiety detection and highlighting the benefits of multimodal learning in mental health assessment.

As a workshop-stage publication, the study lacks extensive validation in real-world clinical environments. Although performance improvements were observed, the reported F1-scores remained relatively modest, suggesting that the framework may still be susceptible to false positives and missed classifications when deployed outside controlled experimental settings.

### **2.2 AI-Powered Mental Health Assessment Using Speech and Text Analysis**

Patil and Faratkhane (2025) proposed a dual-channel mental health screening system that processes speech and text separately before combining them through multimodal fusion. The textual component utilizes transformer-based language

understanding to analyse semantic and sentiment patterns, while the acoustic component extracts speech characteristics such as pitch, jitter, shimmer, and MFCCs to capture behavioural cues associated with mental health conditions. By integrating both channels, the framework provides a scalable and non-invasive approach for depression and anxiety screening.

A major limitation of the study is the absence of validation in real healthcare environments. The framework relies heavily on conventional fusion strategies and does not address how performance may be affected by noisy audio, inaccurate transcriptions, diverse accents, or age-related vocal variations. As a result, its practical reliability remains largely unverified.

### **2.3 Mental Health Monitoring from Speech and Language**

Zubiaga et al. (2022) developed a mental health monitoring framework within the European MENHIR project that combines lightweight acoustic representations with self-supervised text embeddings to classify anxiety and depression in realistic conversational environments. The study also explored the role of emotional states as secondary indicators, demonstrating that tracking affective changes alongside speech and language patterns can improve the distinction between temporary emotional reactions and persistent mental health conditions.

The study provides limited information regarding dataset size, participant demographics, and diagnostic validation procedures. Furthermore, because the text representations were trained primarily on written language corpora, the system may experience performance degradation when handling fragmented speech, pauses, slang, and other characteristics common in spontaneous conversations.

### **2.4 Multimodal and Multiresolution Depression Detection from Speech and Facial Landmark Features**

Nasir et al. (2016) proposed a multimodal framework for estimating depression severity using speech and facial behaviour extracted from the DAIC-WOZ dataset. The authors utilized acoustic representations, speaker-level characteristics, and

facial landmark dynamics across multiple temporal resolutions to capture long-term behavioural patterns associated with depression. Their results demonstrated that analysing behavioural signals over different time scales can improve depression recognition and reduce errors caused by short-term emotional fluctuations.

The framework was heavily optimized for a benchmark dataset and may not generalize effectively to real-world clinical environments. In addition, the approach depends on reliable facial landmark tracking and controlled recording conditions, which may be difficult to maintain in practical deployment scenarios.

## **2.5 Leveraging Audio and Text Modalities in Mental Health: A Study of LLM Performance**

Ali et al. (2024) investigated the effectiveness of modern multimodal Large Language Models for mental health screening using both audio and textual information. Using the E-DAIC dataset, the authors evaluated several commercial language models under text-only, audio-only, and combined audio-text settings. Their findings showed that multimodal inputs consistently outperformed unimodal approaches, demonstrating the potential of large language models to analyse conversational content alongside vocal characteristics for depression and PTSD assessment.

A significant limitation of the study is its dependence on proprietary commercial models and prompt-based inference. The framework is highly sensitive to prompt design, lacks transparency regarding internal decision-making processes, and has not been validated beyond a single benchmark dataset, limiting its generalizability and reproducibility.

## **2.6 When LLMs Meet Acoustic Landmarks: Integrating Speech into Large Language Models for Depression Detection**

Zhang et al. (2024) proposed a framework that integrates speech-derived acoustic landmarks directly into the input of Large Language Models for depression detection. Instead of relying on computationally expensive audio-text alignment

architectures, the authors transformed acoustic indicators into structured textual representations and combined them with interview transcripts. The approach achieved state-of-the-art performance on the DAIC-WOZ dataset and demonstrated that speech-derived behavioural information can effectively complement textual analysis within language-model-based screening systems.

The framework depends heavily on the accuracy of the acoustic landmark extraction stage. Errors introduced during speech processing, background noise, or poor recording quality can directly propagate into the language model, reducing overall screening accuracy and limiting robustness in uncontrolled environments.

## Chapter 3 Procedure and Methods

The development of the proposed multimodal mental health screening and support system follows a structured approach that combines speech and text analysis to identify potential indicators of mental well-being. Recent studies have shown that integrating multiple communication modalities can improve mental health assessment by capturing complementary information from both linguistic and behavioral signals (Nasir et al., 2016; Alosbhan et al., 2021). The proposed system is designed to process user communication through multiple input modalities and generate meaningful screening results along with supportive feedback.

The first stage focuses on data preparation using publicly available mental health datasets containing speech recordings, textual information, and associated assessment labels. The collected data is cleaned, organized, and standardized to ensure consistency throughout the development process. Publicly available benchmark datasets have played an important role in advancing automated mental health screening research and provide a reliable foundation for model development and evaluation (Nasir et al., 2016).

Next, the system processes user input through two parallel pathways. Speech input is converted into textual form while simultaneously preserving important vocal characteristics. These two sources of information provide complementary perspectives on the user's communication and are analyzed independently before being combined. Previous research has demonstrated that linguistic content captures semantic indicators of mental health, while acoustic information provides valuable behavioral cues that may not be apparent from text alone (Alosbhan et al., 2021).

The extracted linguistic and acoustic information is then processed to identify patterns associated with mental health indicators. By considering both what is communicated and how it is communicated, the system aims to provide a more comprehensive assessment than approaches relying on a single modality. Similar multimodal approaches have shown improved performance in identifying



depression, anxiety, and other mental health conditions compared to unimodal systems (Diep et al., 2022; Zubiaga et al., 2022).

After the analysis stage, information from the different modalities is combined to generate a unified representation of the user input. This integrated representation is used to produce mental health screening results and confidence estimates. An explainability component is then utilized to provide understandable insights into the factors contributing to the generated assessment. Explainability has been identified as an important requirement for trustworthy mental health AI systems, particularly in sensitive application domains where users benefit from understanding the reasoning behind system outputs (Zhang et al., 2022).

Finally, a user-facing application is developed to enable practical interaction with the system. Users can either provide speech through a microphone or enter text directly. The system processes the input, performs the analysis, and presents the screening results together with supportive feedback and explanations. Recent advances in Large Language Models have demonstrated their potential for generating supportive, context-aware responses that can enhance user interaction and accessibility (Zhang et al., 2024; Ali et al., 2024).

### **3.1 Simplified Workflow of the Project**

1. User input is obtained either through speech recording or direct text entry.
2. The input data is pre-processed and standardized for further analysis.
3. Linguistic and acoustic information is extracted from the provided input.
4. Information from multiple modalities is combined to generate a unified representation.
5. The system analyses the combined information to identify potential mental health indicators and generate screening results.
6. Explainable insights and supportive feedback are generated and presented to the user through the application interface.

By following this workflow, the system remains modular and flexible. Individual components can be improved, replaced, or extended independently, allowing future

enhancements and supporting additional research in multimodal mental health screening and assistive technologies..

## **Chapter 4. System Requirement Specifications**

This chapter outlines the requirements for developing the proposed multimodal mental health screening and support system. It defines the functionalities the system must provide, the quality attributes it must maintain, and the hardware environment required for development and deployment. These specifications ensure that the system remains practical, reliable, and accessible for its intended users.

### **4.1 Software Specifications**

The software specifications describe the features, behaviour, and performance characteristics required for the successful operation of the system. These requirements are divided into functional requirements, which describe what the system should do, and non-functional requirements, which describe how the system should perform.

#### **4.1.1 Functional Requirements**

1. User Input: The system shall accept user input in the form of either speech recordings or typed text.
2. Speech Processing: The system shall process speech input and convert it into a machine-readable format for further analysis.
3. Mental Health Screening: The system shall analyse user communication and generate a mental health screening result based on identified linguistic and behavioural indicators.
4. Multimodal Analysis: The system shall combine information obtained from speech and text to improve the quality of the screening process.
5. Explainable Results: The system shall provide understandable explanations and indicators that contribute to the generated assessment.

6. Supportive Feedback Generation: The system shall generate supportive and informative feedback based on the screening results.
7. Result Display: The system shall present screening results, explanations, and generated feedback through a clear and user-friendly interface.
8. Evaluation Support: The system shall record prediction results during testing and evaluation phases to facilitate performance analysis using standard metrics.

#### **4.1.2 Non-Functional Requirements**

1. Responsiveness: The system shall process user inputs and generate results within a reasonable time to support interactive use.
2. Reliability: The system shall provide consistent outputs for identical inputs and handle incomplete or invalid user inputs without failure.
3. Usability: The interface shall be simple, intuitive, and understandable for users without technical expertise.
4. Modularity: The system shall be designed in a modular manner so that individual components can be modified or upgraded independently.
5. Privacy: User-provided speech recordings and text inputs shall not be permanently stored without user consent.
6. Maintainability: The software shall be well-structured and documented to support future development, debugging, and enhancement.
7. Scalability: The system architecture shall allow future integration of additional datasets, analysis modules, and support features without major redesign

## 4.2 Hardware Specifications

- Processor: Intel Core i5 (8th Generation or newer) or equivalent
- RAM: 8 GB minimum; 16 GB recommended for model training and experimentation
- GPU: NVIDIA GPU with CUDA support (recommended for model training and accelerated inference)
- Storage: At least 10 GB of free disk space for datasets, model files, application resources, and project documentation
- Operating System: Windows 10/11 or Linux Ubuntu 20.04 and above
- Audio Hardware: Microphone for voice input and testing of speech-based functionalities

## **Chapter 5. Project Planning and Scheduling**

### **5.1 Requirement Analysis and Planning (Week 1–2)**

- a) Review existing research on mental health screening, speech analysis, text analysis, and multimodal artificial intelligence systems to understand current approaches and challenges.
- b) Study publicly available mental health datasets and determine suitable data sources for system development, training, and evaluation.
- c) Finalize the overall system architecture, analysis workflow, screening methodology, and evaluation strategy for the proposed system.

### **5.2 System Design (Week 3–4)**

- a) Design the end-to-end system pipeline, including user input, data processing, multimodal analysis, screening, explainability, and feedback generation components.
- b) Define the interaction between speech processing, text analysis, multimodal fusion, and response generation modules within the system.
- c) Establish evaluation criteria, performance metrics, and testing procedures for assessing the effectiveness of the proposed approach.

### **5.3 Model Development (Week 5–7)**

- a) Implement the data preprocessing pipeline and prepare speech and textual data for analysis.
- b) Develop and configure the analysis modules responsible for extracting meaningful information from user communication.
- c) Train, validate, and refine the screening model using selected datasets and optimize its performance based on evaluation results.

## 5.4 Integration and Interface Development (Week 7–9)

- a) Develop the application workflow that supports both speech and text-based user interaction.
- b) Build a user-friendly interface that accepts user inputs, displays screening results, and presents explanations and supportive feedback.
- c) Integrate all system components into a unified platform and conduct preliminary testing to ensure smooth operation and compatibility.

## 5.5 Testing and Evaluation (Week 10–11)

- a) Evaluate the screening performance of the system using standard metrics such as accuracy, precision, recall, F1-score, and other relevant measures.
- b) Assess the usability, responsiveness, and reliability of the application under different usage scenarios and input conditions.
- c) Identify limitations, address performance issues, conduct final system validation, and prepare project documentation and reports for submission.

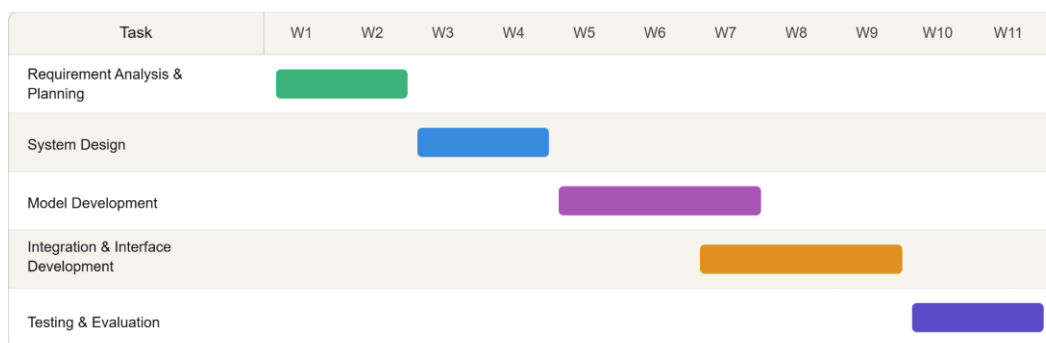


Figure 1: Gantt Chart

## References

- Ali, A. A., Fouda, A. E., Hanafy, R. J., & Fouda, M. E. (2024). Leveraging audio and text modalities in mental health: A study of LLMs performance. arXiv. <https://arxiv.org/abs/2412.10417>
- Aloshban, N., Esposito, A., & Vinciarelli, A. (2021). What you say or how you say it? Depression detection through joint modeling of linguistic and acoustic aspects of speech. *Cognitive Computation*.
- Diep, B., Stanojevic, M., & Novikova, J. (2022). Multi-modal deep learning system for depression and anxiety detection. arXiv. <https://arxiv.org/abs/2212.00969>
- Nasir, M., Jati, A., Shivakumar, P. G., Chakravarthula, S. N., & Georgiou, P. (2016). Multimodal and multiresolution depression detection from speech and facial landmark features. In *Proceedings of the 6th International Audio/Visual Emotion Challenge and Workshop (AVEC 2016)*.
- Patil, S., & Faratkhane, S. (2025). AI-powered mental health assessment using speech and text analysis. *International Journal for Multidisciplinary Research*, 7(2). <https://doi.org/10.36948/ijfmr.2025.v07i02.41988>
- Zhang, T., Schoene, A. M., Ji, S., & Ananiadou, S. (2022). Natural language processing applied to mental illness detection: A narrative review. *npj Digital Medicine*, 5, 46.
- Zhang, X., Liu, H., Xu, K., Zhang, Q., Liu, D., Ahmed, B., & Epps, J. (2024). When LLMs meet acoustic landmarks: An efficient approach to integrate speech into large language models for depression detection. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Zubiaga, I., Menchaca, I., de Velasco, M., & Justo, R. (2022). Mental health monitoring from speech and language. In *Proceedings of the Workshop on Speech, Music and Mind (SMM 2022)*